

Title

AI-Powered Intelligent Candidate Ranking System

Team Name:

Prime Coders

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Diagram

AI Hiring System

Job Description



AI Understanding



Candidate Ranking

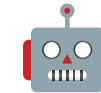


Best Candidates

Use icons:



Job →



AI →



Resume →

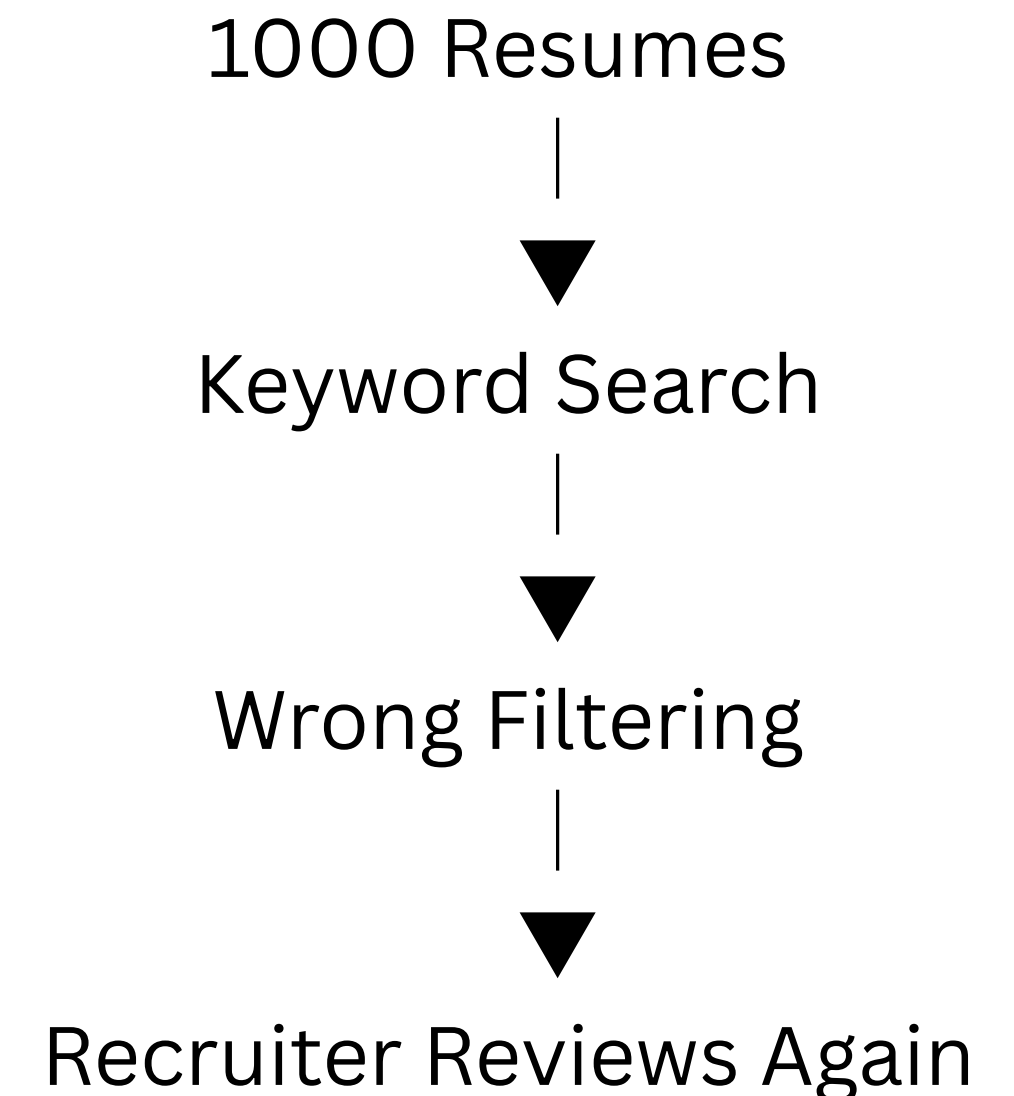


Ranking

Problem Statement

- Existing Recruitment Process
- Recruiters receive hundreds of resumes for every job opening.
- Traditional Applicant Tracking Systems (ATS) rely mainly on keyword matching, often overlooking qualified candidates whose resumes use different terminology. This results in:
- Missing highly suitable candidates
- Large manual screening effort
- Time-consuming recruitment
- Biased or inconsistent selection

Diagram



Proposed Solution

Our Solution

- We built an AI-powered candidate ranking system that understands the meaning of both the job description and candidate profiles instead of relying only on keyword matching.
- Our solution:
- Reads complete job descriptions
- Understands candidate profiles semantically
- Calculates multiple matching scores
- Produces an explainable ranked shortlist

Diagram

AI Ranking Engine

Job Description



Semantic Embedding

Candidate Profiles



Similarity Matching



Hybrid Scoring



Top Candidates

Dataset

Input Data

Our system uses two primary datasets.

Job Dataset

Contains:

- Job Title
- Required Skills
- Experience
- Responsibilities
- Education

Candidate Dataset

Contains:

- Resume
- Skills
- Projects
- Experience
- Education
- Certifications
- Platform Activity

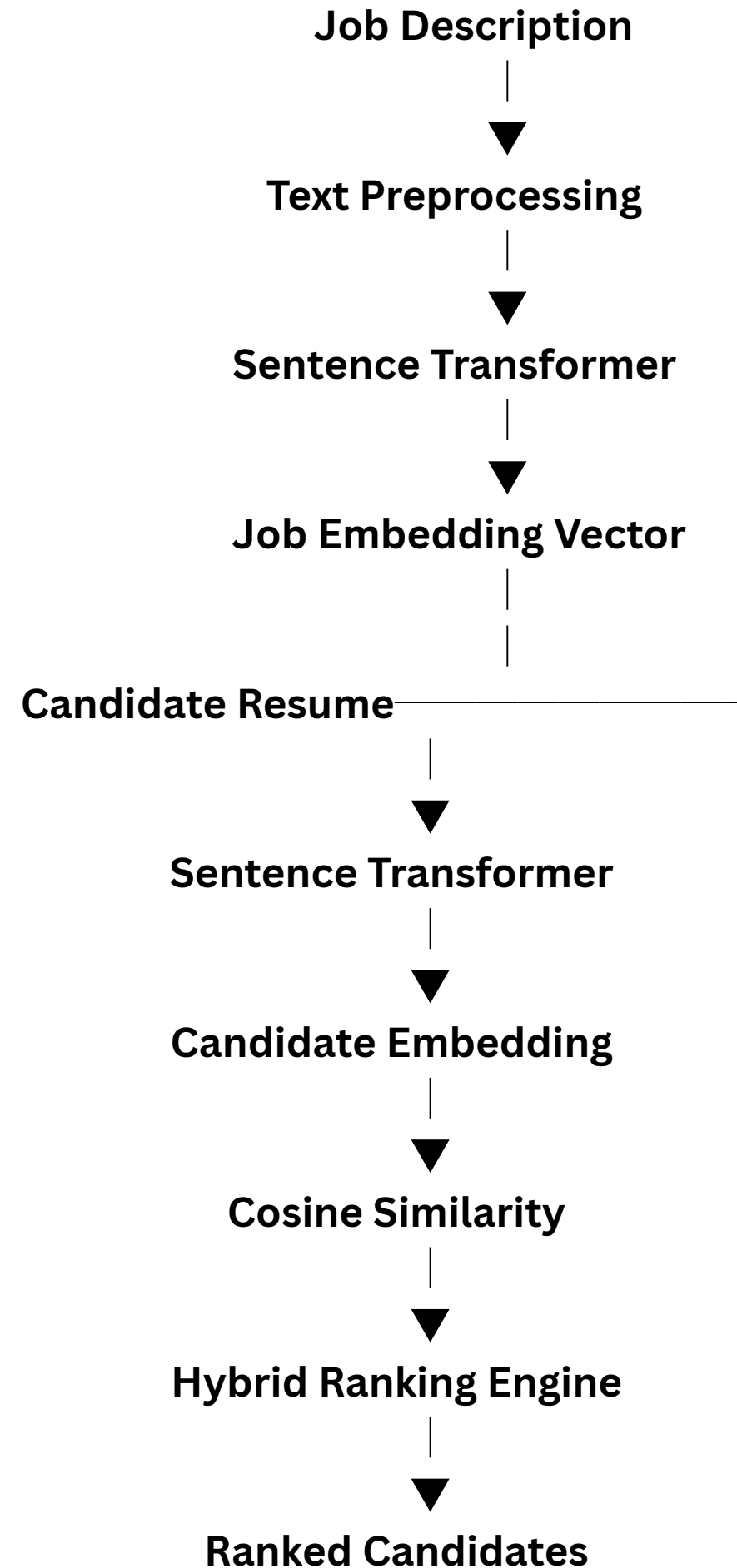
Diagram

Dataset



AI Processing

System Architecture



Components

- Text Cleaning
- Embedding Generation
- Semantic Search
- Feature Scoring
- Final Ranking

Methodology

Diagram

Cleaning



Embedding



Similarity



Hybrid Score



Ranking

Step 1

- Text Preprocessing
- Lowercase conversion
- Stopword removal
- Cleaning special characters

Step 2

- Embedding Generation
- Sentence Transformer converts text into dense vectors.

Step 3

- Semantic Matching
- Cosine Similarity measures how closely a candidate matches the job description.

Step 4

- Hybrid Scoring
- Additional factors are incorporated:
 - Skills
 - Experience
 - Education
 - Activity Score

Step 5

- Final Ranking
- Candidates are sorted according to the final score

AI Model

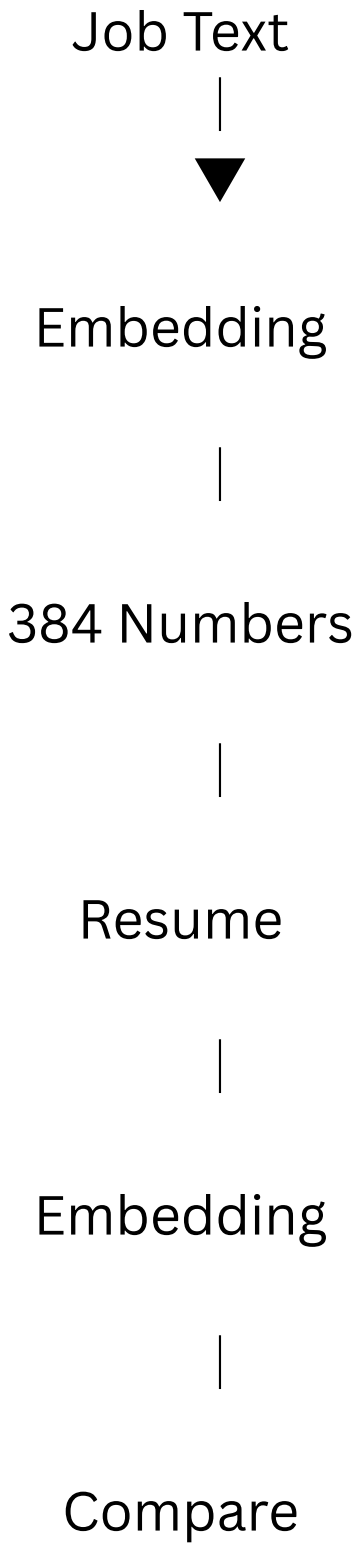
Sentence Transformer

Model Used:

all-MiniLM-L6-v2

- Reason for Selection
- Fast inference
- Lightweight
- High semantic accuracy
- Generates 384-dimensional embeddings
- Suitable for semantic searc

Diagram



Ranking Algorithm

Formula:

Final Score

=

0.50 × Semantic Similarity

+

0.20 × Skill Match

+

0.15 × Experience

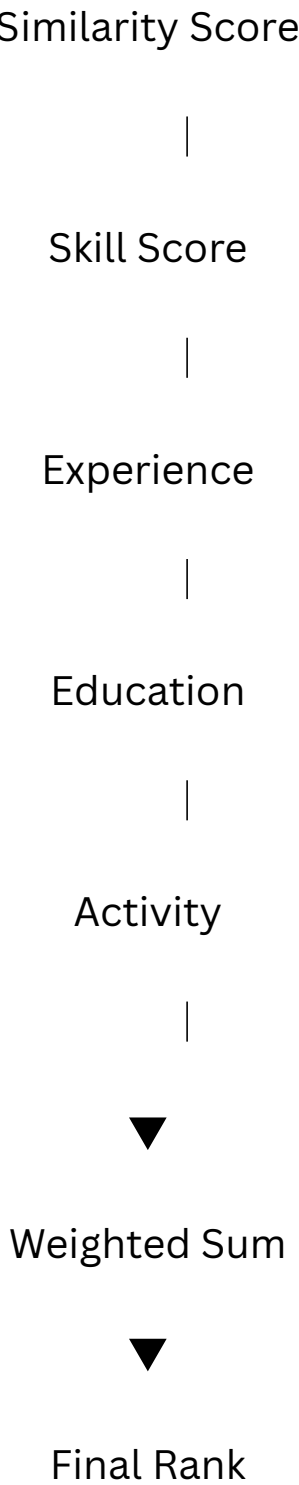
+

0.10 × Activity

+

0.05 × Education

Flowchart:



Explainable AI

Instead of showing only a score, our system provides reasons behind every recommendation.

- Example
- Candidate ID: 205
- Overall Score: 94%
- Reason:
- Python Expert
- Machine Learning Experience
- AWS Certified
- 4 Years Experience
- Resume closely matches job description

Diagram:

AI Decision



Explain Score



Recruiter Trust

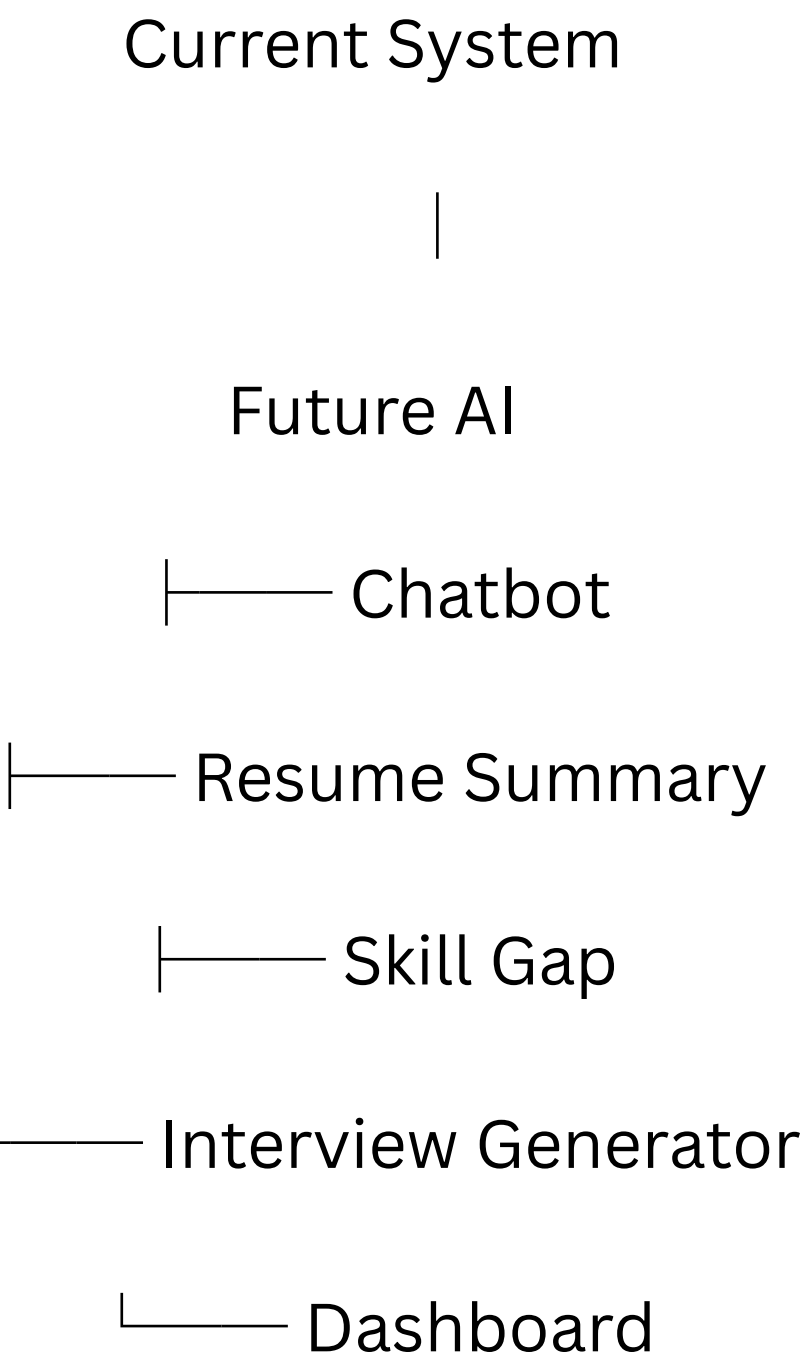
Results:

Rank	Candidate	Score
1	Alice	94%
2	Rahul	91%
3	John	89%
4	Sara	87%

Future Enhancements

Diagram

- Future improvements include:
- LLM-based interview question generation
 - Resume summarization
 - Bias detection
 - Recruiter dashboard
 - Skill gap analysis
 - Multi-language resume support
 - Interview scheduling assistant



Conclusion

- Achievements
- Our system successfully:
- Understands job descriptions semantically
- Matches candidates beyond keyword search
- Produces explainable rankings
- Reduces recruiter effort
- Improves hiring quality
- Supports scalable AI-assisted recruitment

Final Diagram:

Job Description

AI Ranking Engine

Best Candidate

Happy Recruiter